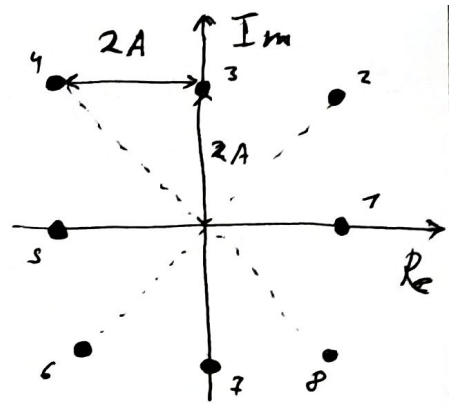


① a) $|s_i - s_{i+1}| = 2A$; $N = 8$

$$P(s_i) = P(s_j) = \frac{1}{N} = \frac{1}{8} \quad \forall i, j$$

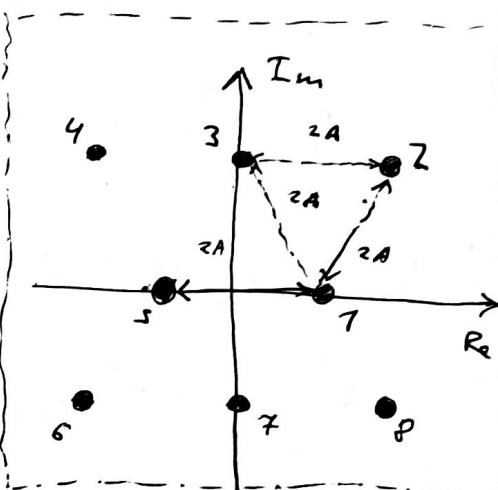
$$\langle E_s \rangle = \sum_{i=1}^N |s_i|^2 P(s_i) = \frac{1}{8} \left(\sum_{i=1}^4 |s_{2i}|^2 + \right.$$

$$\left. + \sum_{i=1}^3 |s_{2i+1}|^2 \right) = \frac{1}{8} (4 \cdot 8A^2 + 4 \cdot 4A^2) = A^2 6$$



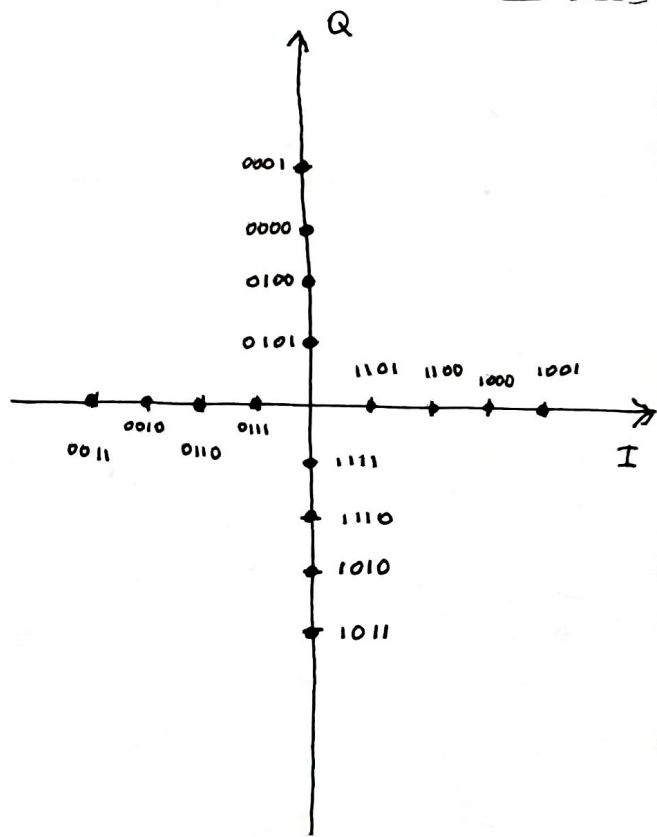
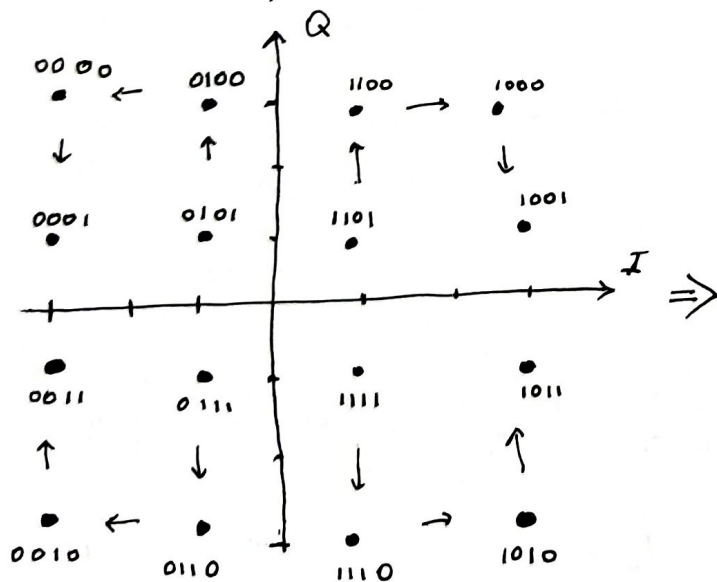
b) $\langle E_s \rangle = \frac{1}{8} \cdot \sum_{i=1}^8 |s_i|^2 = \frac{1}{8} \left((|s_1|^2 + |s_5|^2) + (|s_3|^2 + |s_7|^2) + (|s_2|^2 + |s_4|^2 + |s_6|^2 + |s_8|^2) \right) =$

$$= \frac{1}{8} (2 \cdot A^2 + 2 \cdot 3A^2 + 4 \cdot 7A^2) = 4.5 A^2$$



Constellation (b) is more power-efficient than constellation (a)

② We know, QAM 16:



③ Frequency-modulated signal with memory.
Continuous phase with partial response CPM.

$$h = \frac{1}{2}; \quad g(t) = \begin{cases} \frac{1}{4T}, & 0 \leq t \leq 2T \\ 0, & \text{otherwise} \end{cases}$$

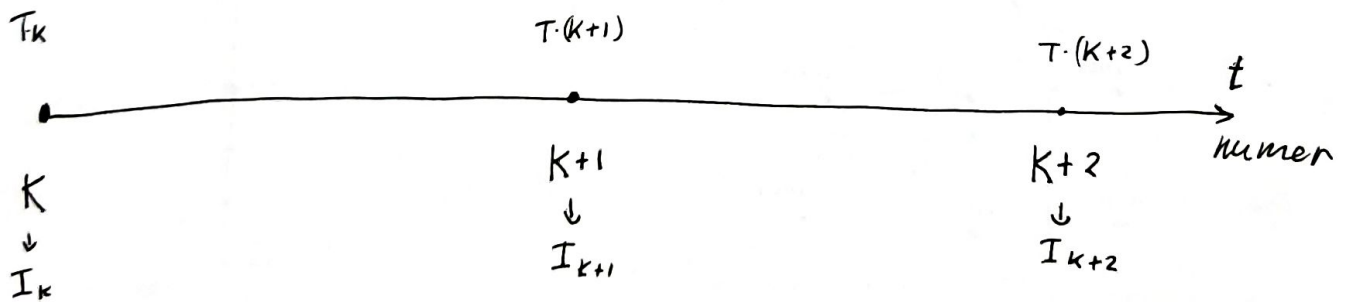
Passband signal: $s(t) = \sqrt{\frac{2E}{T}} \cos(2\pi f_c t + \phi(t; \vec{I}) + \phi_0)$

$$\begin{aligned} \phi(t; \vec{I}) &= 4\pi T f_d \int_{-\infty}^t \sum_{k=-\infty}^n I_k g(\tau - kT) d\tau = \begin{cases} q(t) = \int_0^t g(\tau) d\tau \\ 2\pi h \sum_{k=-\infty}^n I_k q(t - kT), & nT \leq t \leq (n+1)T \end{cases} \\ & \quad \left. \begin{array}{l} \\ \end{array} \right\} h = 2 f_d T \end{aligned}$$

$\{I_k\}; I_k \in \{1, -1\} \leftarrow W < 0 \leq$ for plotting phase-trellis

$\& I_k \in \{-(M-1), \dots, -1, 1, 2, \dots, M-1\}$

$\exists$ current stage number = K , Accumulated phase = ϕ_0 .
 $\phi(t; \vec{I}) = \pi \sum_{k=0}^n I_k q(t - kT)$

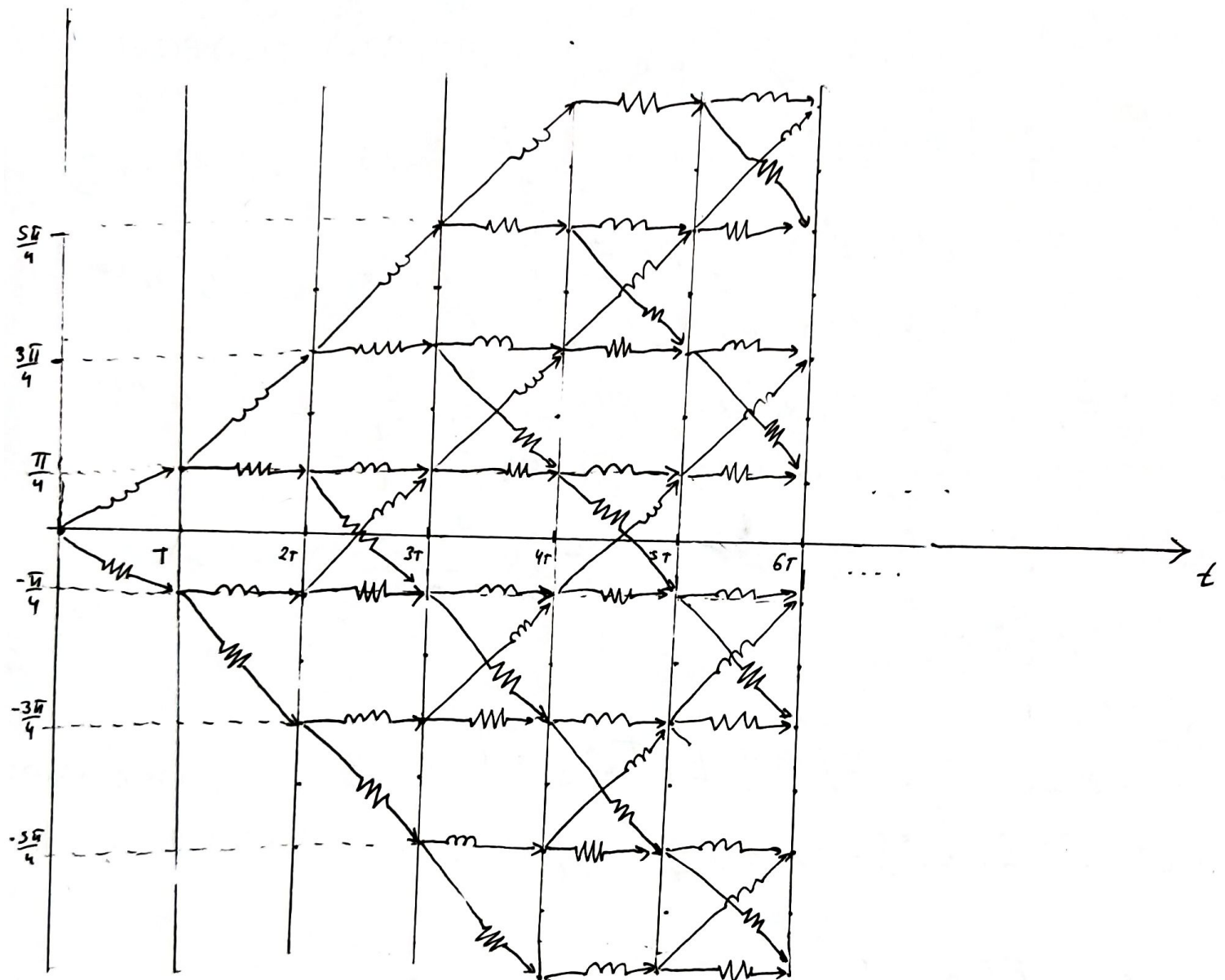


$$\phi_K = \phi_0 \quad \phi_{K+1} = \phi_0 + \pi \left[\frac{I_{K-1}}{4} + \frac{I_K}{4} \right] \quad \phi_{K+2} = \phi_{K+1} + \pi \left[\frac{I_K}{4} + \frac{I_{K+1}}{4} \right]$$

$$\phi_2 = \phi_{2-1} + \pi \left[\frac{I_{2-2}}{4} + \frac{I_{2-1}}{4} \right], \quad \phi_0 = 0.$$

• phase-trellis $I_k \in \{1, -1\}$

$\text{---} \rightsquigarrow \Rightarrow I_k = 1$; $\text{---} \rightsquigarrow \Rightarrow I_k = -1$



④ Phase-modulated signal

$$u(t) = \sum_n I_n g(t - nT) ; I_n \in \left\{ \frac{1+i}{\sqrt{2}}, \frac{1-i}{\sqrt{2}}, \frac{-1+i}{\sqrt{2}}, \frac{-1-i}{\sqrt{2}} \right\}$$

$$P(I_n) = P(I_k) = \frac{1}{4} \quad \forall n, k \in \mathbb{N}; \quad P(I_n \cap I_k) = P(I_n) P(I_k)$$

1) $S_u(f) = \frac{1}{T} |G(f)|^2 S_I(f) = \frac{1}{T} \sum_{k=-\infty}^{+\infty} G_k(f) \exp(-i2\pi k f T)$

$$G_k(f) = E[S_c(f; \vec{I}_k) S_c^*(f; \vec{I}_0)], \quad S_c(f; \vec{I}_k) = F[S_c(t; \vec{I}_k)]$$

$$S_c(t; \vec{I}_k) = \sqrt{I_k} g(t - kT) = I_k g(t - kT)$$

$$G_k(f) = E[I_k \cdot I_0^* F[g(t - kT)] F^*[g(t - kT)]] = E[I_k \cdot I_0^*] \cdot G(f) \cdot$$

$$\exp(-i2\pi f kT) \cdot G^*(f) \exp(i2\pi f kT) = R_I(k) |G(f)|^2 \Rightarrow$$

$$\Rightarrow S_u(f) = \frac{1}{T} \sum_{k=-\infty}^{+\infty} G_k(f) \exp(-i2\pi k f T) = \frac{|G(f)|^2}{T} \underbrace{\sum_{k=-\infty}^{+\infty} R_I(k) \exp(-i2\pi k f T)}_{S_I(f)}$$

$$G(f) = F[g(t)]$$

2) $S_I(f) = ?$

$$R_I(k) = E[I_k \cdot I_0^*] = E[I_k] \cdot E[I_0^*] = 0, \quad k \neq 0 \Rightarrow R_I(k) = \delta(k) \Rightarrow$$

$$R_I(0) = E[I_0 \cdot I_0^*] = \frac{1}{4} \cdot 1 + \frac{1}{4} \cdot 1 + \frac{1}{4} \cdot 1 + \frac{1}{4} \cdot 1 = 1 \Rightarrow S_I(f) = 1$$

3) $G_a(f) = F[g_a(t)] ; g_a(t) = \begin{cases} A, & 0 \leq t \leq T \\ 0, & \text{otherwise} \end{cases} \Rightarrow$

$$G_a(f) = \int_0^T A e^{-i2\pi f t} dt = T A e^{-\pi i f T} \text{sinc}(\pi f T) \Rightarrow$$

$$\Rightarrow S_{u_a}(f) = \frac{(TA)^2 \text{sinc}^2(\pi f T)}{T} = T A^2 \text{sinc}^2(\pi f T)$$

$$G_b(f) = F[g_b(t)]; \quad g_b(t) = \begin{cases} A \sin\left(\frac{\pi t}{T}\right), & 0 \leq t \leq T \\ 0, & \text{otherwise} \end{cases}$$

$$G_b(f) = \int_0^T A \left(\frac{e^{i\frac{\pi t}{T}} - e^{-i\frac{\pi t}{T}}}{2i} \right) e^{-2\pi i f t} dt =$$

$$\int_0^T e^{it\left(\frac{\pi}{T} - 2\pi f\right)} dt = T e^{i\frac{T}{2}\left(\frac{\pi}{T} - 2\pi f\right)} \text{sinc}\left(\frac{T}{2}\left(\frac{\pi}{T} - 2\pi f\right)\right) \Rightarrow$$

$$-\int_0^T e^{it\left(-\frac{\pi}{T} - 2\pi f\right)} dt = -T e^{-i\frac{T}{2}\left(\frac{\pi}{T} + 2\pi f\right)} \text{sinc}\left(\frac{T}{2}\left(\frac{\pi}{T} + 2\pi f\right)\right)$$

$$\Rightarrow G_b(f) = \frac{AT}{2i} \cdot e^{-i\pi T f} \cdot i \left[\text{sinc}\left(\frac{\pi}{2} - \pi T f\right) + \text{sinc}\left(\frac{\pi}{2} + \pi T f\right) \right] =$$

$$= \frac{AT}{2} e^{-i\pi T f} \left[\text{sinc}\left(\frac{\pi}{2} - \pi T f\right) + \text{sinc}\left(\frac{\pi}{2} + \pi T f\right) \right]$$

$$S_{u_b}(f) = \frac{A^2 T}{4} \left[\text{sinc}\left(\frac{\pi}{2} - \pi T f\right) + \text{sinc}\left(\frac{\pi}{2} + \pi T f\right) \right]^2.$$

4) The PSD of S_{u_b} decays faster than the PSD of S_{u_a} . As a result, out-of-band emissions are lower, but the main lobe becomes wider.